

Marketing Campaign Analysis — Phase 1: Exploratory Data Analysis

Omar Touzani · Datasets: marketing_campaign.csv (2,240 customers) + Amazon Reviews (28,332 reviews, 65 products)

I explored both datasets to understand customer behaviour, campaign response patterns, and review sentiment. The two datasets share no join key — the Amazon reviews are used purely as a standalone NLP exercise in Phase 5 and cannot be tied to specific campaign customers.

Headline numbers

Average age	45 years	Top spending category	Wines (\$680,816)
Average income	\$52,247	Top purchase channel	Store (12,970 buys)
Average total spend	\$606	Best campaign	Latest — 14.91%
Customers with children	1,602 (71.5%)	Avg review rating	4.51 / 5

Key findings

Wines dominate spending. Wines alone account for \$680,816 — almost twice the next category (Meat at \$373,968) and more than the bottom four categories combined. Full ranking: Wines, Meat, Gold, Fish, Sweets, Fruits.

Store is the dominant channel, but web has a conversion gap. 12,970 store purchases vs 9,150 web. Customers visit the website 5.3 times per month on average — high relative to purchases. Worth exploring in Phase 3.

The latest campaign far outperforms the previous five. 14.91% acceptance vs 1.34–7.46% for campaigns 1–5. This is also the target variable for Phase 4 propensity modelling.

Reviews are heavily positive-skewed. 90.2% positive (4–5 stars), 4.3% neutral, 5.6% negative. Phase 5 will need class weighting or resampling, with evaluation by confusion matrix → classification report → AUC-ROC.

Data quality issues and Phase 2 actions

Issue	Detail	Action
Constant columns	Z_CostContact, Z_Revenue (std = 0)	Drop both
Age outliers	3 customers with Year_Birth ≤ 1900	Drop the 3 rows
Income outlier	1 customer with Income = \$666,666	Drop the row
Missing income	24 rows (1.07%) — biased low (4.2% response vs 15%)	Drop, flag bias
Marital status junk	8 categories incl. YOLO, Absurd, Alone	Consolidate into Single
Reviews duplication	1,669 reviews share (product, text)	De-duplicate
Very short reviews	2,538 reviews under 20 characters	Filter from sentiment

What's next

Phase 2 — Data Cleaning and PostgreSQL loading. Apply the actions above, consolidate marital status categories, and load the cleaned tables to the project database for Phases 3–6.